Token accuracy is a sufficient statistic for deduction

56 cells, N=48 fall ensemble, leak-free (--shuffle-adj) throughout

THE PUZZLE THAT STARTED IT

Stage C's first cells train a FULL transformer at d_model 128 to train NLL 1.62 — essentially where ATTENTION-ONLY at d_model 256 sits, 1.63 — and it scores mean solve 0.027 against attention-only's 0.304. Eleven-fold, at the same training loss. Every 'did it train?' hedge here is pinned to a train-NLL threshold (1.64, finding 44), so that hedge does not order these models. Validation TOKEN ACCURACY does: 0.459 against 0.465–0.483, with the full d256 hub at 0.494.

THE QUESTION

Once you know a cell's token accuracy, does its ARCHITECTURE tell you anything more about whether it solves? If no, no cut here destroys a 'deduction circuit' — each only slides the model along a curve DEPTH already traces. If some cut sits BELOW the curve, that one breaks something solve-specific and is the interesting cell.

THE TEST

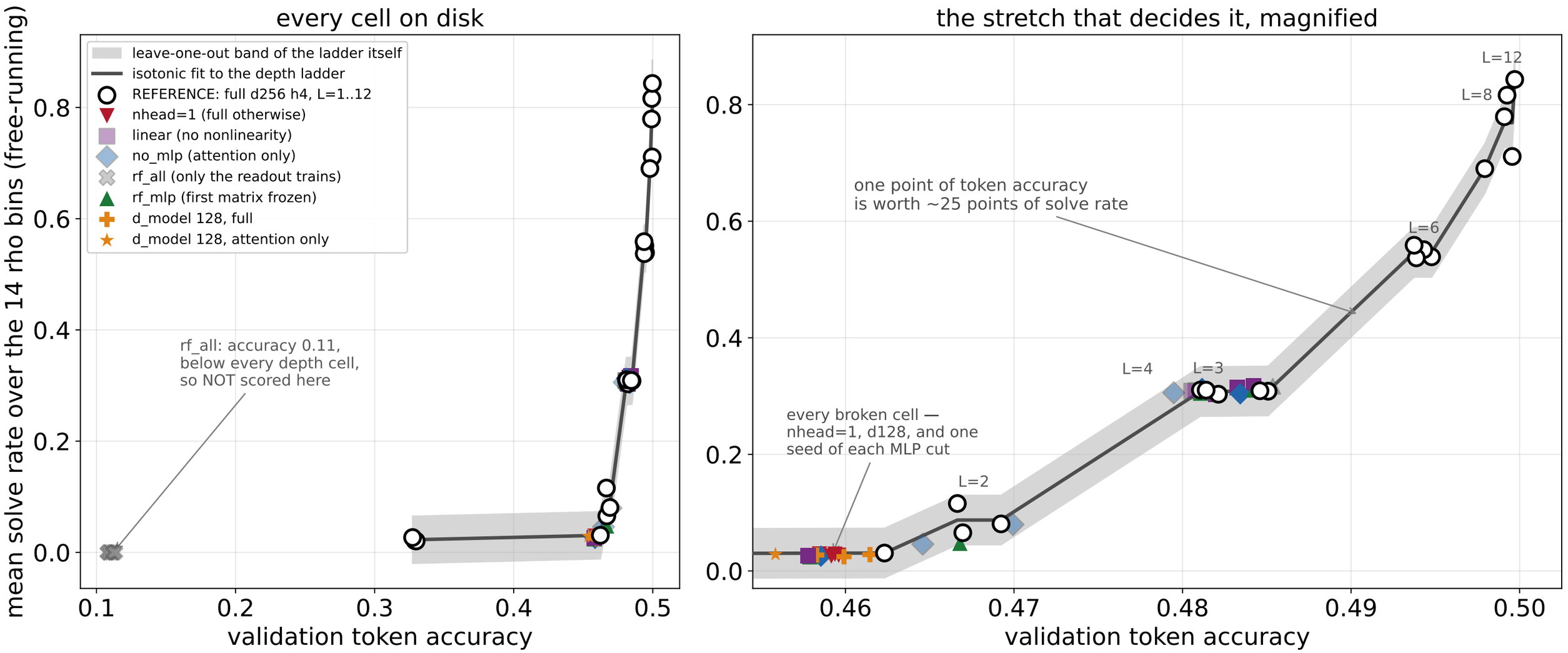
The reference is the DEPTH LADDER ALONE — full, d256, 4 heads, L = 1..12, 21 cells — the only axis spanning a wide accuracy range (0.327 to 0.500) with the architecture intact. The fit is ISOTONIC: the one thing assumed is that more accurate models do not solve less often, since assuming a shape would settle the question by choosing a functional form. Everything else is HELD OUT, and the bar is the reference's own leave-one-out residual, RMS 0.0423.

THE ANSWER: NO — ACCURACY IS ENOUGH

All 23 well-determined held-out cells land on the curve, over 10 rungs: L=4 h=1 full, L=4 h=4 linear, L=4 h=4 no_mlp, L=4 h=4 rf_mlp, L=6 h=1 full, L=6 h=4 full d128, L=6 h=4 linear, L=6 h=4 no_mlp, L=6 h=4 no_mlp d128, L=6 h=4 rf_mlp. 7 are DISCRIMINATING — the curve sends them above 0.10, so agreement is a constraint and not two numbers near zero — and those cover EVERY MLP cut at both depths. Worst miss 0.0091 against the bar 0.0423. So 'the MLP is all-or-nothing' gets a mechanism: the three cuts are indistinguishable because they land at the same accuracy, 0.481–0.485, where the curve is near-vertical — one accuracy point is worth twenty-five of solve rate. Weakest defensible form: model quality here is ONE-DIMENSIONAL.

Correlational: nothing here was intervened on. It is also a statement about this grid's accuracy range, which no cell except rf_all leaves. NOT the trivial compounding law — independent errors at accuracy 0.48 over a trace this long would put every solve rate at zero, so these models' errors are concentrated on positions the solve metric forgives.

Every architecture lands on the curve that DEPTH alone traces



Hollow circles are the reference and define the curve; every other marker was HELD OUT of the fit. Faded markers are cells whose bracketing depth points are further apart than the bar — reported, but not what the verdict rests on. The curve is near-vertical between 0.47 and 0.50, and THAT steepness, not the architecture, is why rho_50 censors and why three seeds of one rung can disagree by 0.28 in mean solve.

The seven cells that could have falsified it

held-out cells the curve sends above 0.10, where landing on it is a real constraint

architecture	seed	token acc	predicted	observed	miss
L=6 h=4 rf_mlp	0	0.4842	0.3086	0.3104	+0.0017
L=6 h=4 linear	2	0.4842	0.3086	0.3177	+0.0091
L=6 h=4 no_mlp	2	0.4834	0.3084	0.3039	-0.0044
L=6 h=4 linear	1	0.4833	0.3083	0.3146	+0.0063
L=4 h=4 linear	2	0.4820	0.3079	0.3029	-0.0051
L=4 h=4 rf_mlp	1	0.4811	0.3079	0.3050	-0.0029
L=6 h=4 no_mlp	0	0.4812	0.3079	0.3130	+0.0051

worst miss 0.0091 against the bar 0.0423 → none of them is off the curve

```
ALL 35 HELD-OUT CELLS, BY HOW WELL THE CURVE PINS THEM
  firm    23 cells, 10 rungs — bracketed by depth cells closer together than the bar. 7 of these are the table above.
         worst rung mean residual: L=6  h=4 rf_mlp -0.0190, still inside 0.0423.
  weak     6 cells — they land in the ladder's empty accuracy stretch, where the curve is flat and near zero. Agreement there is
         consistency, not evidence, so they are segregated rather than pooled into the verdict.
  out      6 cells — all rf_all, at token accuracy 0.11. No depth cell is anywhere near that, so they are reported as out of range, never scored.

WHY THE REFERENCE IS THE DEPTH LADDER AND NOTHING ELSE
  It is the only axis that moves accuracy a long way with the architecture intact, and the cells under test must not help draw the curve they
  are judged against — the same non-circularity rule that made the hub's own seed spread, not the pooled s*, the bar for the MLP cuts.
```

56 cells across all three scan dirs; every one N=48, degree 5, 4 colours, fall ensemble, --shuffle-adj, 240000x50 steps. scripts/acc_vs_solve.py at commit c1cfabc. Token accuracy is the final-epoch validation figure already in every history.json — nothing was retrained and no checkpoint was reloaded to produce this deck.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND        N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND        N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND        N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall. TOKEN ACCURACY is how often the model's next token matches this trace under teacher forcing; SOLVE RATE is whether it can generate the whole thing unaided. This deck is about the relation between the two.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

What follows, and what does not

consequences of a one-dimensional quality axis for the minimal-architecture search

1. IT REPLACES THE METRIC THE PROGRAMME WAS STUCK ON

rho_50 censors — eight of the grid's rungs have at least one seed that never crosses 0.5, so their median is a bound and the depth gaps had to be read on mean solve instead. Token accuracy never censors, is already recorded for every cell that has ever been trained here, and predicts solve rate to within 0.009 on the cells where that is a real constraint. Future rungs should be scored on it FIRST and on rho_50 only to confirm.

2. IT DISSOLVES THE TRAINABILITY CAVEAT INTO A NUMBER

Every stripped rung carried the hedge 'part of this cost is that one seed never left the 1.64 plateau'. That hedge can now be quantified instead of repeated: each cut costs one to three points of token accuracy, and because the curve is near-vertical there, that IS the whole solve-rate gap. Nothing is left over for a cut to have broken separately.

3. IT MAKES THE nhead=1 VERDICT SAYABLE

N5 was the one confounded verdict in the grid: one head scored censored 3/3 with 0/3 plateau escapes, so 'worse' could not be separated from 'did not train'. It lands at token accuracy 0.459, on the curve, exactly where full-at-d128 also lands. That still does not say one head CANNOT represent deduction — but it says one head fails the way a too-small model fails, not in some head-specific way. Stage C's warmup arm is still the intervention that settles it.

4. WHAT IT DOES NOT SHOW

Nothing here was intervened on: this is a curve through cells that happen to exist, and an architecture could still be built that sits off it. rf_all is outside the range entirely and was excluded, not explained. And a sufficient statistic for the SOLVE METRIC is not a sufficient statistic for the MECHANISM — two models at equal accuracy can still deduce differently, which is the question the attention-only frontier was chosen to make answerable.

THE FRONTIER THIS LEAVES

Attention-only at L=6: 1,746,490 trainable parameters, rho_50 0.763, token accuracy 0.481, cheaper than full L=3 at the same score. On the matched-budget reading it beats full L=2 outright, 0.304 to 0.081, at 1.746M against 1.741M. It is also the most analytically tractable object in the grid — with no MLP, the residual stream is written only by attention.

Previous best on the same eval: the 11-02 scan, whose matched L=8 cells replicate to within s* (-0.1062). There is no previous result to compare this page against — the accuracy-to-solve curve had not been drawn before, which is why it is published as a method change rather than as an improvement in any score.